

ZOOKEEPER 이슈 재현

김성현

이슈 목록

ZOOKEEPER-2247

디스크 read-only 등으로 쓰기 실패 후 정상으로 돌아오더라도,
리더가 섰다운 되지 않음.
이에 재선출도 일어나지 않고,
트랜잭션 로그 쓰기도 불가능한 버그

ZOOKEEPER-2592

zookeeper 실행 상태에서 디스크 용량이 가득 찼을 때 발생한 응답
불능이 디스크 용량이 다시 확보된 상황에서도 지속되는 버그

ZOOKEEPER-3496

트랜잭션 로그 요청은 1MB까지만 허용함. 즉 1MB 찌음은 요청이 가능.
그러나 이후 내부적으로 실제로 트랜잭션 로그를 쓸 때는 추가적인 내용
들이 붙어 더 큰 크기로 저장됨.
이때 1024Bytes를 넘으면, IOException이 발생하며 시작 불가능한 버그

ZOOKEEPER-2247

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2026-08-14 21:19:02,912 [myid:3] - ERROR [SyncThread:3:ZooKeeperCriticalThread@49] - Severe unrecoverable error, from thread : SyncThread:3
java.io.FileNotFoundException: /datalog/version-2/log.200000001 (Read-only file system)
    at java.io.FileOutputStream.open0(Native Method)
    at java.io.FileOutputStream.open(FileOutputStream.java:270)
    at java.io.FileOutputStream.<init>(FileOutputStream.java:213)
    at java.io.FileOutputStream.<init>(FileOutputStream.java:162)
    at org.apache.zookeeper.server.persistence.FileTxnLog.append(FileTxnLog.java:205)
    at org.apache.zookeeper.server.persistence.FileTxnSnapLog.append(FileTxnSnapLog.java:314)
    at org.apache.zookeeper.server.ZKDatabase.append(ZKDatabase.java:476)
    at org.apache.zookeeper.server.SyncRequestProcessor.run(SyncRequestProcessor.java:140)
2026-08-14 21:19:02,912 [myid:3] - INFO [SyncThread:3:ZooKeeperServer$ZooKeeperServerListenerImpl@446] - Thread SyncThread:3 exits, error code 1
2026-08-14 21:19:02,912 [myid:3] - INFO [SyncThread:3:ZooKeeperServer@469] - shutting down
2026-08-14 21:19:02,912 [myid:3] - INFO [SyncThread:3:SessionTrackerImpl@225] - Shutting down
2026-08-14 21:19:02,912 [myid:3] - INFO [SyncThread:3:PrepRequestProcessor@767] - Shutting down
2026-08-14 21:19:02,912 [myid:3] - INFO [SyncThread:3:ProposalRequestProcessor@88] - Shutting down
2026-08-14 21:19:02,912 [myid:3] - INFO [SyncThread:3:CommitProcessor@184] - Shutting down
2026-08-14 21:19:02,912 [myid:3] - INFO [SyncThread:3:Leader$ToBeAppliedRequestProcessor@661] - Shutting down
2026-08-14 21:19:02,912 [myid:3] - INFO [ProcessThread(sid:3 cport:-1)::PrepRequestProcessor@143] - PrepRequestProcessor exited loop!
2026-08-14 21:19:02,912 [myid:3] - INFO [CommitProcessor:3:CommitProcessor@153] - CommitProcessor exited loop!
2026-08-14 21:19:02,912 [myid:3] - INFO [SyncThread:3:FinalRequestProcessor@415] - shutdown of request processor complete
2026-08-14 21:19:02,913 [myid:3] - INFO [SyncThread:3:SyncRequestProcessor@209] - Shutting down
2026-08-14 21:19:02,934 [myid:3] - DEBUG [LearnerHandler-/172.18.0.2:54672:Leader@584] - Count for zxid: 0x200000001 is 1
2026-08-14 21:19:02,935 [myid:3] - DEBUG [LearnerHandler-/172.18.0.4:38610:Leader@584] - Count for zxid: 0x200000001 is 2
2026-08-14 21:19:04,000 [myid:3] - INFO [SessionTracker:SessionTrackerImpl@162] - SessionTrackerImpl exited loop!
2026-08-14 21:20:05,761 [myid:3] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory@192] - Accepted socket connection from /172.18.0.4:49858
2026-08-14 21:20:05,765 [myid:3] - DEBUG [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:ZooKeeperServer@842] - Session establishment request from client /172.18.0.4:49858
```

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2026-08-14 21:30:40,017 [myid:3] - DEBUG [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:ZooKeeperServer@842] - Session establishment request from client /172.18.0.4:46138
2026-08-14 21:30:40,017 [myid:3] - INFO [NIOServerCxn.Factory:0.0.0.0/0.0.0.0:2181:ZooKeeperServer@900] - Client attempting to establish new session at /172.18.0.4:46
2026-08-15 06:53:44,974 [myid:3] - ERROR [LearnerHandler-/172.18.0.2:54672:LearnerHandler@631] - Unexpected exception causing shutdown while sock still open
java.net.SocketTimeoutException: Read timed out
    at java.net.SocketInputStream.socketRead0(Native Method)
    at java.net.SocketInputStream.socketRead(SocketInputStream.java:116)
    at java.net.SocketInputStream.read(SocketInputStream.java:171)
    at java.net.SocketInputStream.read(SocketInputStream.java:141)
    at java.io.BufferedInputStream.fill(BufferedInputStream.java:246)
    at java.io.BufferedInputStream.read(BufferedInputStream.java:265)
    at java.io.DataInputStream.readInt(DataInputStream.java:387)
    at org.apache.jute.BinaryInputArchive.readInt(BinaryInputArchive.java:63)
    at org.apache.zookeeper.server.quorum.QuorumPacket.deserialize(QuorumPacket.java:83)
    at org.apache.jute.BinaryInputArchive.readRecord(BinaryInputArchive.java:99)
    at org.apache.zookeeper.server.quorum.LearnerHandler.run(LearnerHandler.java:542)
2026-08-15 06:53:44,975 [myid:3] - WARN [LearnerHandler-/172.18.0.2:54672:LearnerHandler@644] - ***** GOODBYE /172.18.0.2:54672 *****
2026-08-15 06:53:44,975 [myid:3] - ERROR [LearnerHandler-/172.18.0.4:38610:LearnerHandler@631] - Unexpected exception causing shutdown while sock still open
java.net.SocketTimeoutException: Read timed out
    at java.net.SocketInputStream.socketRead0(Native Method)
    at java.net.SocketInputStream.socketRead(SocketInputStream.java:116)
    at java.net.SocketInputStream.read(SocketInputStream.java:171)
    at java.net.SocketInputStream.read(SocketInputStream.java:141)
    at java.io.BufferedInputStream.fill(BufferedInputStream.java:246)
    at java.io.BufferedInputStream.read(BufferedInputStream.java:265)
    at java.io.DataInputStream.readInt(DataInputStream.java:387)
    at org.apache.jute.BinaryInputArchive.readInt(BinaryInputArchive.java:63)
```

ZOOKEEPER-2592

```
2026-08-18 05:18:16,742 [myid:] - WARN [NIOWorkerThread-1:WorkerService$$ScheduledWorkRequest@164] - Unexpected exception
java.nio.BufferUnderflowException
    at java.nio.Buffer.nextGetIndex(Buffer.java:510)
    at java.nio.HeapByteBuffer.getLong(HeapByteBuffer.java:427)
    at org.apache.zookeeper.server.NIOServerCnxn.checkFourLetterWord(NIOServerCnxn.java:514)
    at org.apache.zookeeper.server.NIOServerCnxn.readLength(NIOServerCnxn.java:535)
    at org.apache.zookeeper.server.NIOServerCnxn.doIO(NIOServerCnxn.java:331)
    at org.apache.zookeeper.server.NIOServerCnxnFactory$IOWorkRequest.doWork(NIOServerCnxnFactory.java:530)
    at org.apache.zookeeper.server.WorkerService$$ScheduledWorkRequest.run(WorkerService.java:162)
    at java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1149)
    at java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:624)
    at java.lang.Thread.run(Thread.java:750)
2026-08-18 05:18:16,744 [myid:] - INFO [NIOWorkerThread-1:NIOServerCnxn@606] - Closed socket connection for client /0:0:0:0:0:0:0:1:32924 (no session established for client)
2026-08-18 05:18:25,429 [myid:] - INFO [NIOServerCnxnFactory.AcceptThread:0.0.0.0/0.0.0.0:2181:NIOServerCnxnFactory$AcceptThread@296] - Accepted socket connection from /127.0.0.1:42968
2026-08-18 05:18:25,444 [myid:] - DEBUG [NIOWorkerThread-2:ZooKeeperServer@914] - Session establishment request from client /127.0.0.1:42968 client's lastZxid is 0x0
2026-08-18 05:18:25,456 [myid:] - INFO [NIOWorkerThread-2:ZooKeeperServer@964] - Client attempting to establish new session at /127.0.0.1:42968
2026-08-18 05:18:25,459 [myid:] - DEBUG [NIOWorkerThread-2:SessionTrackerImpl@269] - Adding session 0x10000029b510000
2026-08-18 05:18:25,459 [myid:] - TRACE [NIOWorkerThread-2:ZooTrace@71] - SessionTrackerImpl --- Adding session 0x10000029b510000 30000
2026-08-18 05:18:25,463 [myid:] - TRACE [ProcessThread(sid:0 cport:2181)::ZooTrace@90] - :PsessionId:0x10000029b510000 type:createSession cxid:0x0 zxid:0xfffffffffffffffe txnid:0x0
2026-08-18 05:18:25,468 [myid:] - TRACE [ProcessThread(sid:0 cport:2181)::ZooTrace@71] - SessionTrackerImpl --- Existing session 0x10000029b510000 30000
2026-08-18 05:18:25,469 [myid:] - INFO [SyncThread:0:FileTxnLog@200] - Creating new log file: log.1
2026-08-18 05:18:25,470 [myid:] - ERROR [SyncThread:0:ZooKeeperCriticalThread@48] - Severe unrecoverable error, from thread : SyncThread:0
java.io.IOException: No space left on device
    at java.io.FileOutputStream.writeBytes(Native Method)
    at java.io.FileOutputStream.write(FileOutputStream.java:326)
    at java.io.BufferedOutputStream.flushBuffer(BufferedOutputStream.java:82)
    at java.io.BufferedOutputStream.flush(BufferedOutputStream.java:140)
    at org.apache.zookeeper.server.persistence.FileTxnLog.append(FileTxnLog.java:212)
    at org.apache.zookeeper.server.persistence.FileTxnSnapLog.append(FileTxnSnapLog.java:372)
    at org.apache.zookeeper.server.ZKDatabase.append(ZKDatabase.java:548)
    at org.apache.zookeeper.server.SyncRequestProcessor.run(SyncRequestProcessor.java:122)
2026-08-18 05:18:25,470 [myid:] - INFO [SyncThread:0:ZooKeeperServer$ZooKeeperServerListenerImpl@475] - Thread SyncThread:0 exits, error code 1
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ZOOKEEPER-3496 재현하지 못함

Description

Problem:

ZooKeeper server fails to start, logs following error

```
Exception in thread "main" java.io.IOException: Unreasonable length = 1001025
    at org.apache.jute.BinaryInputArchive.checkLength(BinaryInputArchive.java:127)
    at org.apache.jute.BinaryInputArchive.readBuffer(BinaryInputArchive.java:92)
```

기타사항

자바계열의 로깅 소프트웨어의 경우 (log4j, logback)
로그레벨이 TRACE, DEBUG, INFO, WARN, ERROR 존재

주키퍼의 경우, 도커 허브에 없는 버전이 있기도 함.

주키퍼의 경우 분산시스템을 전제하여 만들었기 때문에
종료가 문제가 아닌 경우가 있음
즉 정상작동 하지 않는데 종료도 하지 않는 경우가 더욱 위험